



**Comsats University Islamabad
(Vehari Campus)**

Submitted by: Naveed Abbas

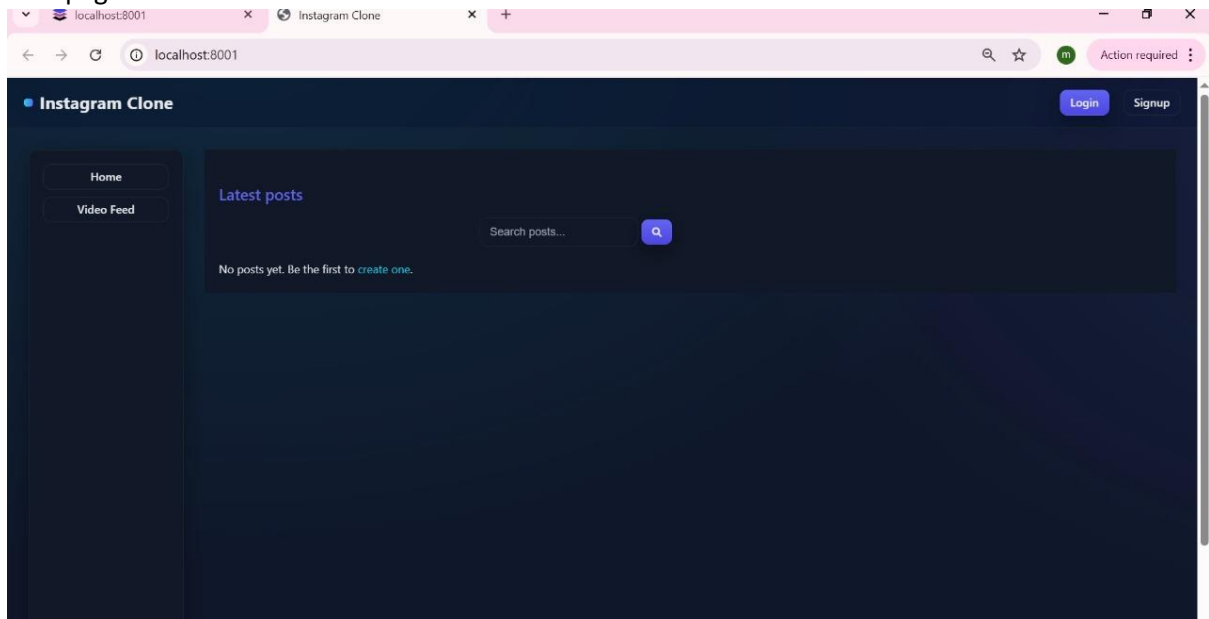
Submitted to: Ma'am Yasmeen Jana

Registration No: FA22-BSE-109

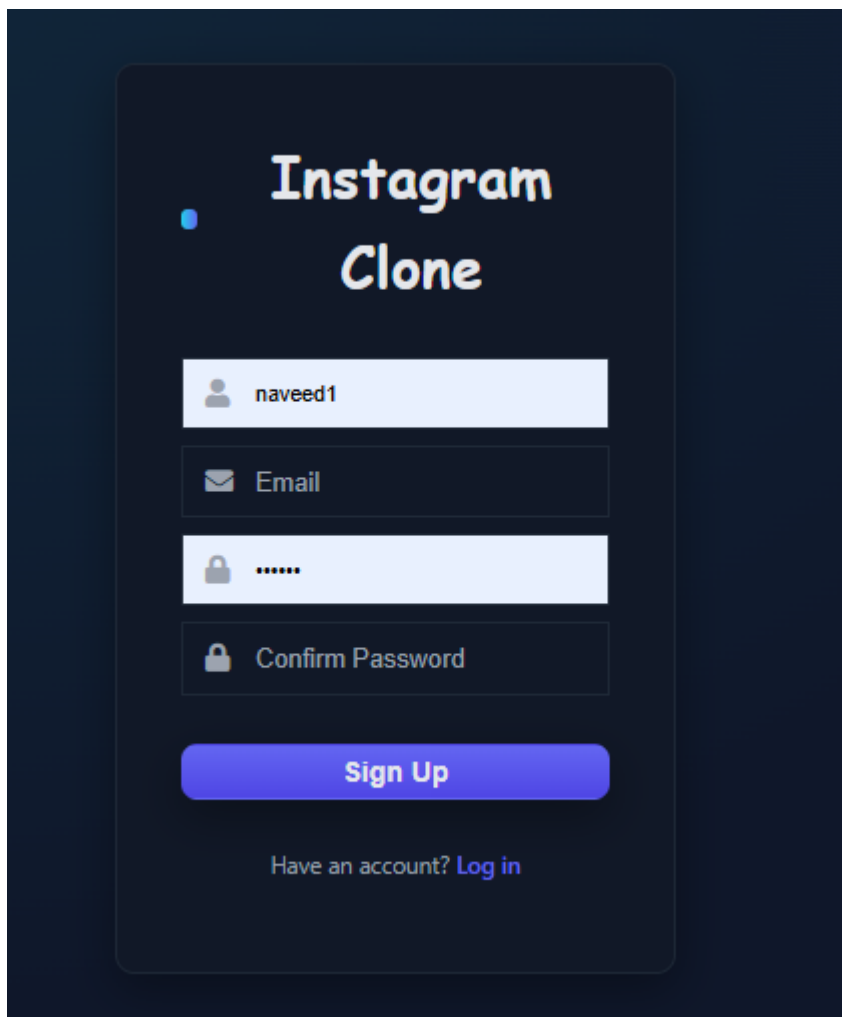
Subject: Advanced web technologies

Topic: Lab Final

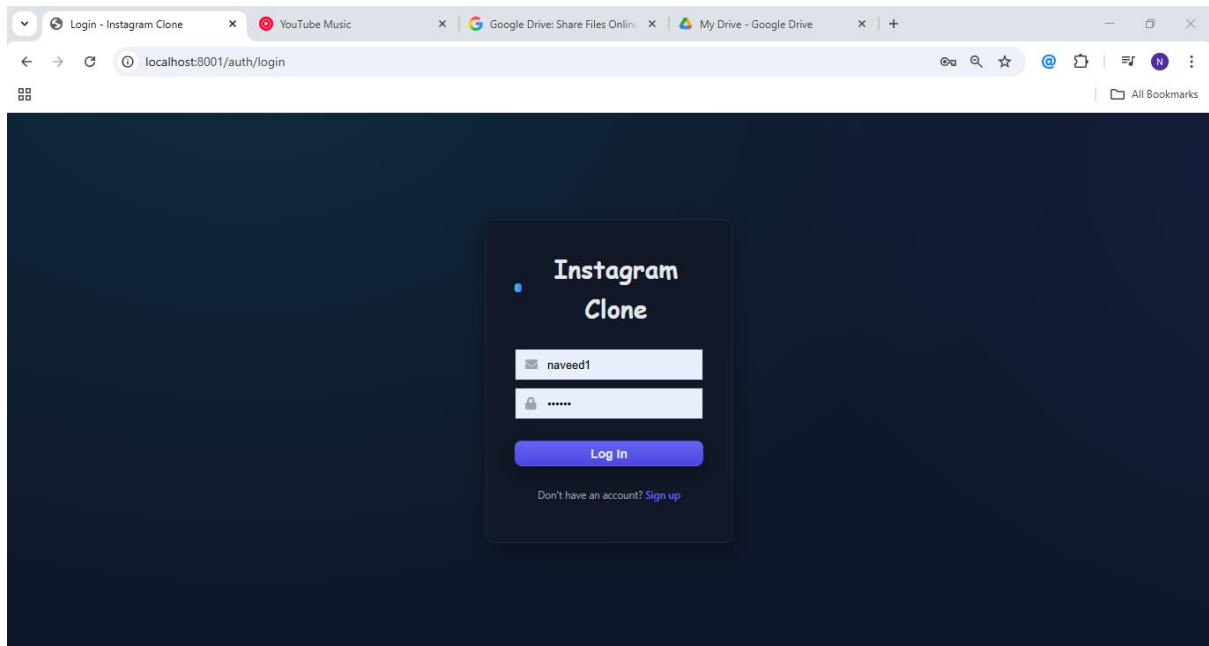
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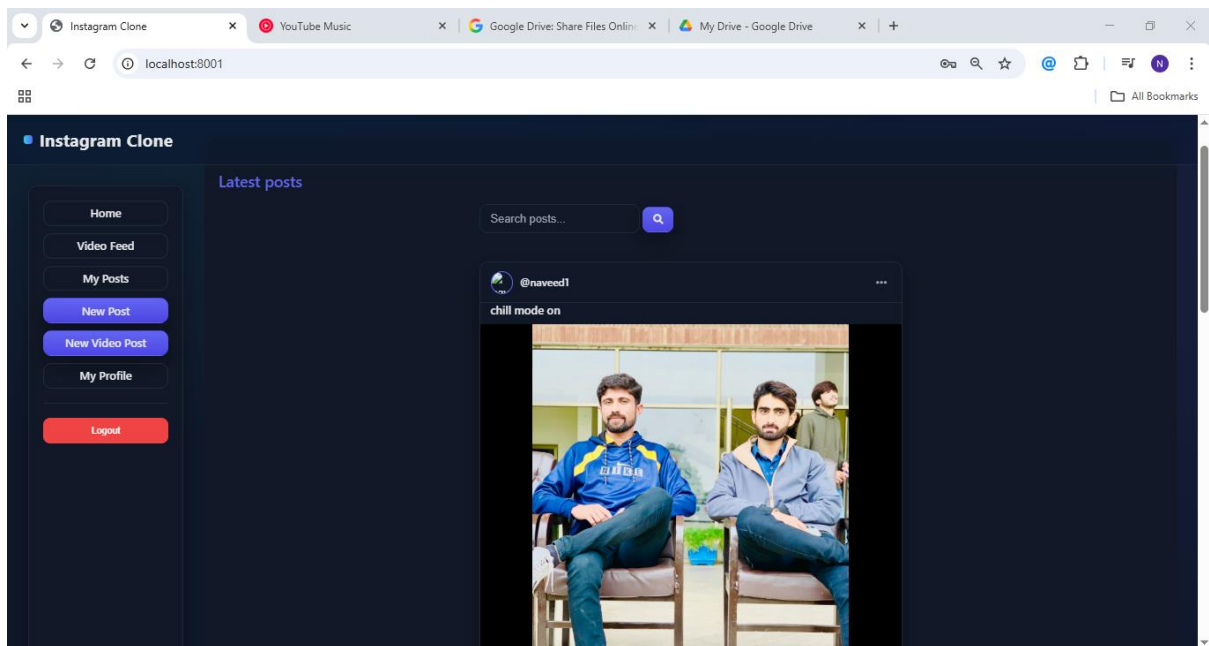
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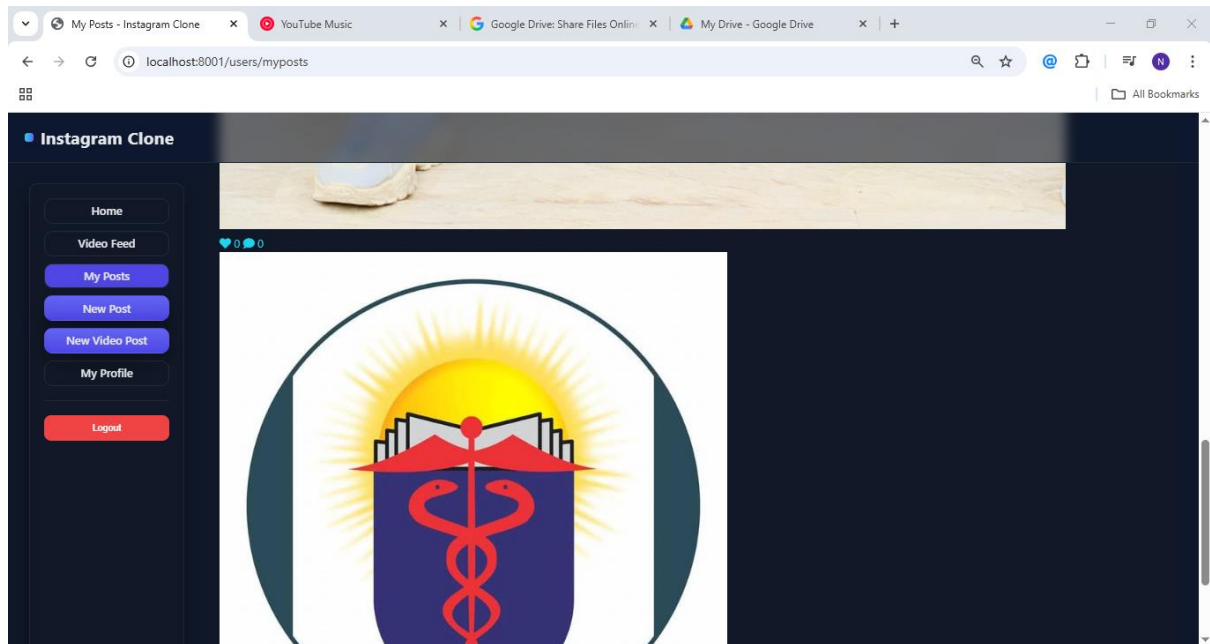
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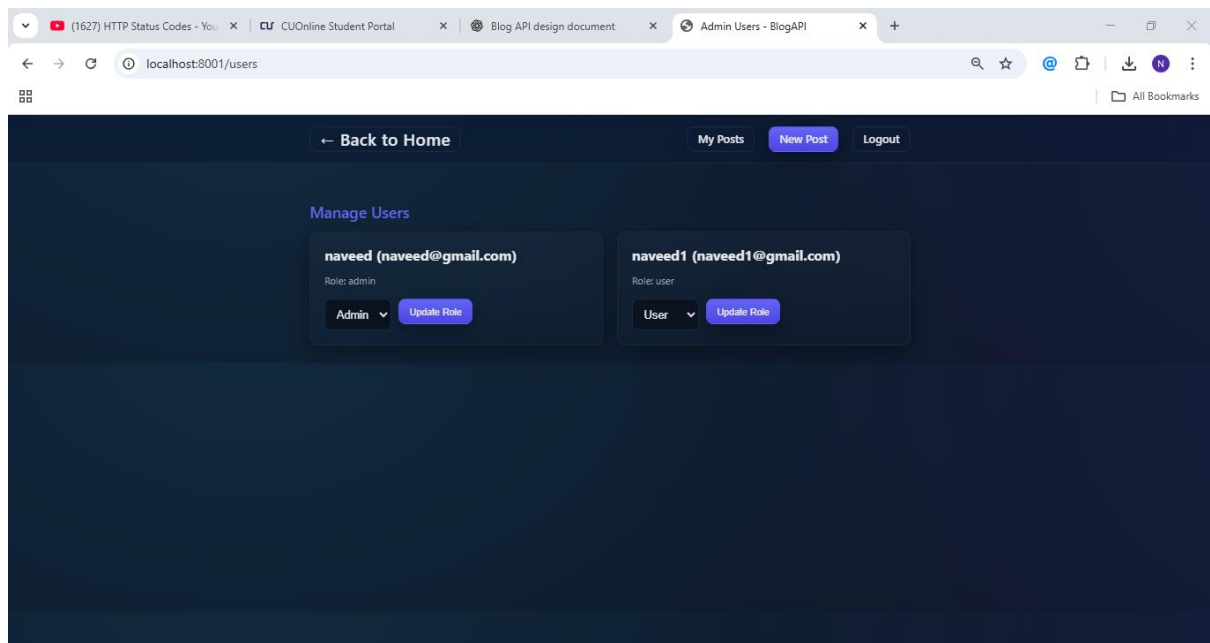
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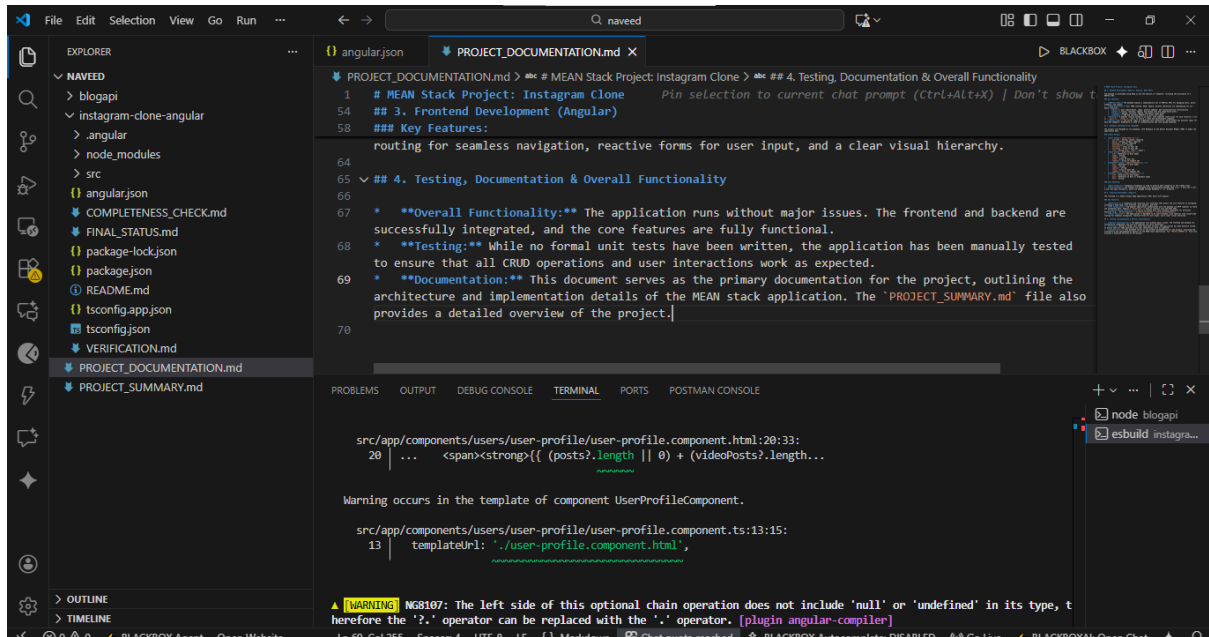
Posts:



Admin management:



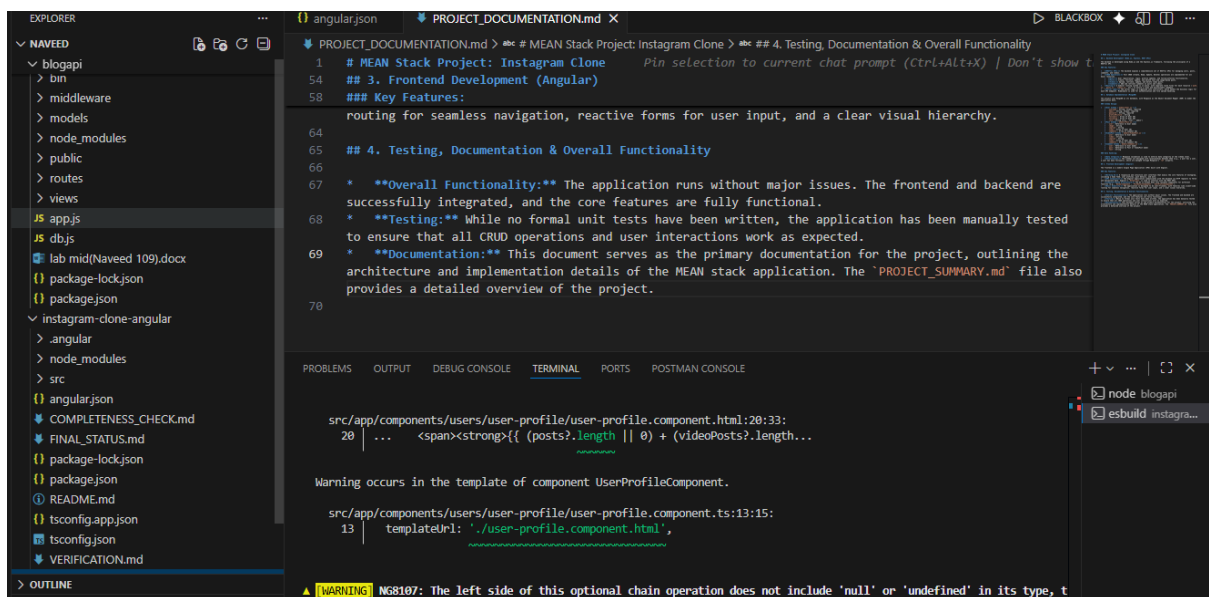
Codes



The screenshot shows the VS Code editor interface. The Explorer sidebar on the left displays the project structure, including files like `PROJECT_DOCUMENTATION.md` and `PROJECT_SUMMARY.md`. The main editor area shows the content of `PROJECT_DOCUMENTATION.md`, which includes sections for project overview, key features, and testing/documentation. The Terminal panel at the bottom displays the output of the build process, showing a warning (NG8107) related to a template URL and a message about the optional chain operator.

```
1 # MEAN Stack Project: Instagram Clone
54 ## 3. Frontend Development (Angular)
58 ### Key Features:
64 routing for seamless navigation, reactive forms for user input, and a clear visual hierarchy.
65 ## 4. Testing, Documentation & Overall Functionality
66
67 * **Overall Functionality:** The application runs without major issues. The frontend and backend are
68 successfully integrated, and the core features are fully functional.
69 * **Testing:** While no formal unit tests have been written, the application has been manually tested
70 to ensure that all CRUD operations and user interactions work as expected.
71 * **Documentation:** This document serves as the primary documentation for the project, outlining the
72 architecture and implementation details of the MEAN stack application. The 'PROJECT_SUMMARY.md' file also
73 provides a detailed overview of the project.
```

```
src/app/components/users/user-profile/user-profile.component.html:20:33:
20 | ... <span><strong>{{ (posts?.length || 0) + (videoPosts?.length...
Warning occurs in the template of component UserProfileComponent.
src/app/components/users/user-profile/user-profile.component.ts:13:15:
13 | templateUrl: './user-profile.component.html',
[WARNING] NG8107: The left side of this optional chain operation does not include 'null' or 'undefined' in its type, therefore the '?.' operator can be replaced with the '.' operator. [plugin angular-compiler]
```



This screenshot is similar to the one above, showing the same VS Code editor interface. The Explorer sidebar now shows a different set of files, including `app.js`, `db.js`, and `package.json`. The main editor area still shows the content of `PROJECT_DOCUMENTATION.md`. The Terminal panel at the bottom displays the same warning (NG8107) and message about the optional chain operator.

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[WARNING] NG8107: The left side of this optional chain operation does not include 'null' or 'undefined' in its type, therefore the '?.' operator can be replaced with the '.' operator. [plugin angular-compiler]
```

Internal model view routes:

MongoDB server:

Collection name	Properties	Storage size	Documents	Avg. document size	Indexes	Total index size
comments	-	20.48 kB	1	130.00 B	1	20.48 kB
posts	-	36.86 kB	2	234.00 B	1	36.86 kB
sessions	-	32.77 kB	5	190.00 B	2	65.54 kB
users	-	36.86 kB	2	359.00 B	3	110.59 kB
videoposts	-	20.48 kB	1	232.00 B	1	20.48 kB

MEAN Stack Project: Instagram Clone

Project Documentation

This document provides a comprehensive overview of the Instagram-like clone project, covering both the backend and frontend components.

1. Project Overview

This project is a full-stack web application that mimics some of the core functionalities of Instagram. It allows users to register, log in, create and share image and video posts, comment on posts, like/dislike posts, and follow/unfollow other users. It also includes an admin role with privileges to manage users, posts, and comments.

2. Technologies Used

2.1. Backend

- **Framework:** Express.js (Node.js)
- **Database:** MongoDB with Mongoose ODM
- **Authentication:** JWT (JSON Web Tokens) and session management with “express-session” and “connect-mongo”
- **File Uploads:** “express-fileupload” for handling image and video uploads
- **View Engine:** EJS (Embedded JavaScript templates) for server-side rendering of some pages
- **Other Key Dependencies:**
 - “bcryptjs”: For hashing passwords
 - “cors”: For enabling Cross-Origin Resource Sharing
 - “morgan”: For HTTP request logging
 - “express-validator”: For input validation

2.2. Frontend

- **Framework:** Angular (version 21)
- **HTTP Client:** Angular's built-in “HttpClient” for making API requests to the backend
- **Routing:** Angular Router for navigating between different components
- **State Management:** RxJS (“BehaviorSubject”) and Angular Signals for managing user authentication state

- **Styling: CSS**

3. Backend Architecture

The backend is a monolithic application built with Express.js.

3.1. Database Models

The application uses Mongoose to define the following data models:

- **User:** Represents a user of the application. Stores information like username, email, hashed password, bio, profile picture, role ("user" or "admin"), and lists of followers and following users.
- **Post:** Represents an image post. Contains a title, caption, a reference to the author (User), the image path, and arrays of likes and dislikes.
- **VideoPost:** Represents a video post. Similar to the "Post" model, but stores a video path instead of an image path.
- **Comment:** Represents a comment on a post. Contains the comment content, a reference to the author (User), and a reference to either a "Post" or a "VideoPost".

3.2. API Endpoints

The backend exposes a RESTful API for the frontend to consume. Here's a summary of the main routes:

1. **"/auth"**: Handles user authentication (signup, login, logout).
2. **"/posts"**: Provides CRUD (Create, Read, Update, Delete) operations for image posts, as well as endpoints for liking and disliking posts.
3. **"/videos"**: Provides CRUD operations for video posts, and endpoints for liking and disliking them.
4. **"/comments"**: Allows creating and deleting comments on both image and video posts.
5. **"/users"**:
 - 1.1. Provides user profile information.
 - 1.2. Handles profile updates.
 - 1.3. **Manages following and unfollowing users.**
 - 1.4. **Includes admin-only endpoints for managing users, posts, and comments.**

3.3. Middleware

1. **"auth.js"**: Contains middleware to restrict access to authenticated users ("restrictToAuthenticated") and admin users ("restrictToAdmin").
2. **"imageUpload.js" & "videoUpload.js"**: Middleware for validating uploaded image and video files.
3. **"logger.js"**: Logs incoming requests.

4. Frontend Architecture

The frontend is a single-page application (SPA) built with Angular.

4.1. Components

The application is divided into several components, each responsible for a specific part of the UI:

- 1 "auth": "LoginComponent" and "SignupComponent" for user authentication.**
- 2 "home": "HomeComponent" displays a feed of posts.**
- 3 "posts": Components for creating, viewing, and editing image posts ("PostNewComponent", "PostDetailComponent", "PostEditComponent").**
- 4 "videos": Components for the video feed, creating, viewing, and editing video posts ("VideoFeedComponent", "VideoNewComponent", "VideoDetailComponent", "VideoEditComponent").**
- 5 "users": Components for user profiles ("UserProfileComponent", "UserProfileEditComponent") and viewing a user's own posts ("MyPostsComponent").**
- 6 "admin": Components for admin functionalities like managing users, posts, and comments ("AdminUsersComponent", "AdminPostsComponent", "AdminCommentsComponent", "AdminUserPostsComponent", "AdminUserCommentsComponent").**
- 7 "shared": Reusable components like the navbar and sidebar.**

4.2. Services

Services are used to encapsulate business logic and data access, making the code more modular and reusable.

- 1 - **“AuthService”**: Manages user authentication state, including login, signup, and logout. It uses RxJS and Angular Signals to provide the current user's authentication status and role to other parts of the application.
- 2 **“PostsService”**: Handles all HTTP requests related to image posts.
- 3 **“VideosService”**: Handles all HTTP requests related to video posts.
- 4 **“CommentsService”**: Manages creating and deleting comments.
- 5 **“UsersService”**: Interacts with user-related API endpoints, such as fetching profiles, updating profiles, and admin actions.

4.3. Routing and Guards

- 1 **“app.routes.ts”**: Defines the application's routes, mapping URLs to their corresponding components.
- 2 **“auth.guard.ts”**: Protects routes that require an authenticated user.
- 3 **“admin.guard.ts”**: Protects routes that are only accessible to admin users.

5. How to Run the Project

5.1. Backend

1. Navigate to the “main” directory.
2. Install dependencies: “npm install”
3. Make sure you have a MongoDB server running locally on “mongodb://127.0.0.1:27017”.
4. Start the server: “npm start”
5. The backend will be running on “http://localhost:8001”.

5.2. Frontend

1. Navigate to the “instagram-clone-angular” directory.
2. Install dependencies: “npm install”
3. Start the development server: “ng serve”
4. Open your browser and go to “http://localhost:4200”.

4. Testing, Documentation & Overall Functionality

Overall Functionality: The application runs without major issues. The frontend and backend are successfully integrated, and the core features are fully functional.

Testing: While no formal unit tests have been written, the application has been manually tested to ensure that all CRUD operations and user interactions work as expected.

Documentation: This document serves as the primary documentation for the project, outlining the architecture and implementation details of the MEAN stack application.